Correlates of peptide-HLA manufacturing success, TCR capture and neoTCR trafficking from patients using the PACTImmune™ Database

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Introduction

PACT Pharma uses a state-of-the-art approach to predict and validate neoepitopes (neoEs) and their cognate T cell receptors (neoTCRs) by capturing neoE-specific T cells from peripheral blood. This neoTCR discovery and validation process is being applied in a clinical trial (NCT03970382) evaluating personalized neoTCR-T cell therapy in solid tumors. The PACTImmune Database (PIDB) accumulates extensive pre-, on- and post-treatment data related to this trial. Here we present PACTImmune Database data on the prediction of neoatnigens, peptide-HLA manufacturing success, success of autologous TCR capture and neoTCR T cell tumor trafficking, all of which have direct applications for prospective patient enrollment and treatment.

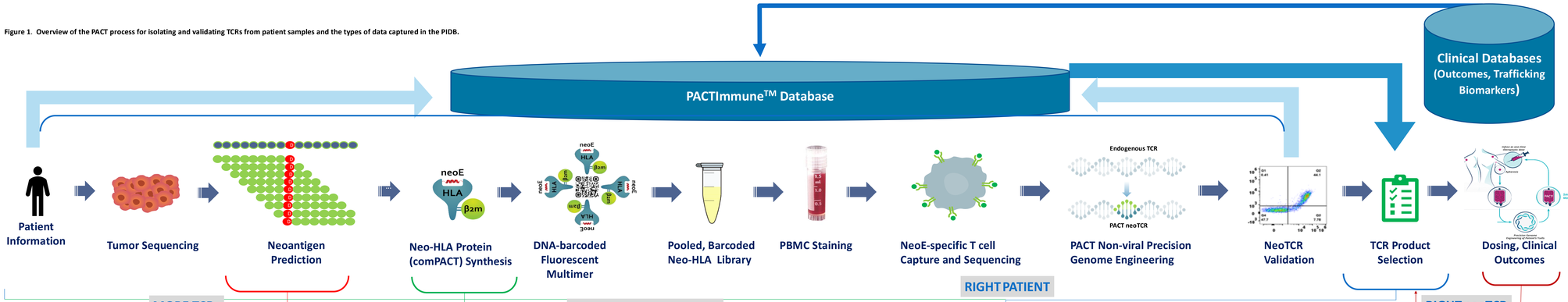
Conclusion

- The PACTImmune Database represents a significant & maturing dataset for patient-specific tumor immunogenicity in solid cancers and provides opportunities to optimize personalized neoTCR T cell treatment.
- The PACTImmune Database of neoE predictions, neoTCR validations and clinical data represents a significant & growing dataset for patient-specific tumor immunogenicity in solid cancers.
- PACTImmune Database-based machine learning improved predictions of neoE-HLAs and translates to a significant increase in successful manufacturing of comPACT molecules.
- The increase in successfully manufactured comPACT molecules is expected to result in an increase in neoE-specific CD8+ T cell capture and actionable TCRs per patient.
- Future improvements in heuristics for epitope selection & prioritization are underway.

Figure 10. Improved comPACT production success rates from ML is predicted to lead to more TCR options per patient. Plot shows retrospective PIDB data of the number of functional TCRs determined by IFN-γ expression verses the number of successful comPACTs made for that patient. A significant positive correlation (P=0.022) is found between the two.

Methods and Results

Figure 1. Overview of the PACT process for isolating and validating TCRs from patient samples and the types of data captured in the PIDB.



PACT uses machine learning for neoantigen predictions powered by the PIDB to make significant improvements over current state of the art approaches

Methods

PACT has developed capabilities for high-throughput manufacturing of single polypeptide (comPACT protein) which consists of the predicted neoE peptide together with beta-2-microglobulin and the HLA heavy chain. comPACT molecules are considered successfully produced when protein yields reach concentrations >1uM. We applied ML to learn patterns that are predictive of neoE-HLAs that can be successfully produced as comPACTs. Data used for this study consisted of >26,000 neoE-HLA predictions for 62 different HLA alleles (see Model Training section for details). For each allele, data was first split into **training** and **testing** data using a 70/30 stratified split. We additionally performed experimental prospective validation of the models. To do this, 603 neoE-HLAs (from 7 previously unseen cancer samples) were selected using netMHCpan4.1 and the newly trained models and comPACT success rate was compared. Performance measured using the area under the receiver operating characteristic curve (AUROC). More successful predictions correlates with more TCRs.

PIDB Neo-HLA Data

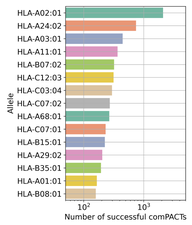


Figure 1. Top 15 HLA alleles with the highest number of successful comPACTs.

PACT ML Outperforms netMHCpan4.1 Elution Rank & Binding Affinity Predictions

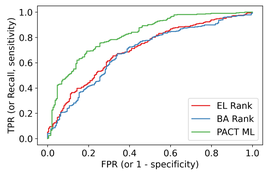


Figure 2. Comparison of ROC curves for 3 neoE-HLA ranking strategies, i.e., elution rank (netMHCpan4.1), binding affinity rank (netMHCpan4.1) and PACT ML scores, for HLA-A02:01.

comPACT success rates were on average 22% higher during prospective validation

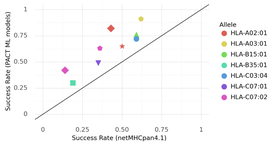


Figure 3. Comparison of the comPACT production success rates when the neoE-HLAs were selected using netMHCpan4.1 and the ML models based on PACTImmune Database. Each dot in the plot denotes a combination of a patient sample (shape) and an HLA allele (color).

Improved Prediction of comPACT Success Rates Allow for Better QC and Patient Stratification

Methods

We used the PIDB to predict manufacturing success rates of our single chain trimer (comPACT) which consists of the predicted neoE peptide together with B2M and the HLA heavy chain. Using multiple linear regression (MLR), we built a model predicting success rates of comPACT protein manufacturing (defined as comPACT protein yields >1uM) and compared it to prediction based solely on expressed tumor mutation burden (eTMB).

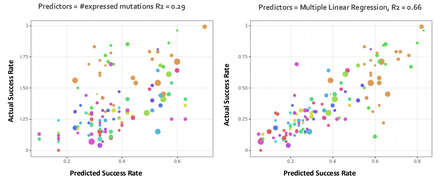


Figure 4. Performance of the selected models on test data (30% of data was previously unseen) in terms of area under the ROC curve (AUROC).

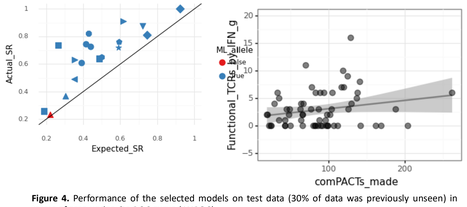


Figure 4. Performance of the selected models on test data (30% of data was previously unseen) in terms of area under the ROC curve (AUROC).

Identifying Which Samples Have the Highest Probability of Success of Yielding Actionable neoTCRs

Methods

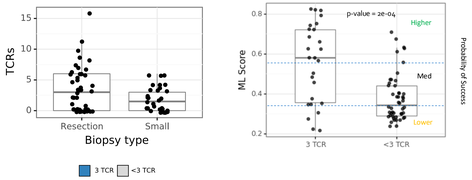
Clinical
- Biopsy site
- Biopsy type
- Liver met?
- Age
- Cancer indication

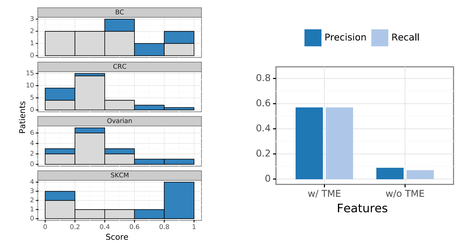
Protein Science
- comPACTs made
- comPACTs > 1.5uM
- Mutations covered in comPACTs > 1.5uM

Bioinformatics
- TMB
- TME data

QC
- PBMC Viability
- Total viable CD8 cells for imPACT
- CD4/CD8 Ratio

Figure 4. We used machine learning leveraging clinical, protein science, bioinformatics and qc characteristics to predict which samples were most likely to yield a multi-neoTCR product.





All Dose Level 1 Patients Examined Show Trafficking of Product to Post Dosing Biopsies and Help Identify TCR Characteristics Predictive of Trafficking

Methods

We evaluated if PIDB neoantigen and TCR characteristics correlate with neoTCR T cell trafficking in patients (n=4) dosed with neoTCR T DL 1 cells. Trafficking was determined by sequencing (neoTCR specific reads) and in some cases, IHC/RNAScope imaging of post-dosing biopsies.

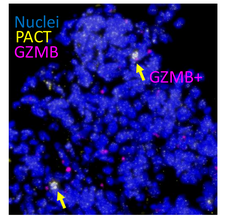


Figure 2. (left) PACTImmune Database summary **(right)** Top 15 HLA alleles with the highest number of successful comPACTs.

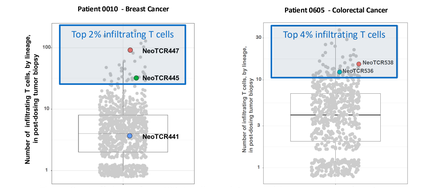


Figure 2. Trafficking of neoTCR T cells as determined by sequencing of post-dosing biopsies shows strong trafficking of specific neoTCRs.

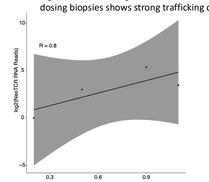


Figure 2. Trafficking of neoTCR T cells as determined by imaging showed a positive correlation with sequencing based approaches.

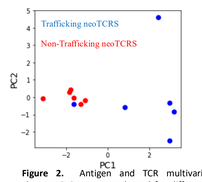


Figure 2. Antigen and TCR multivariate characteristics were evaluated for differences in neoTCRs that traffic to the tumor (blue) verses those that don't (red). PC1 is driven primarily from TCR based characteristics.